

Skill 1 & Skill 2

SKILL 1:

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

```
harshika@Harshika:~/SystemForge_OSSP$ cd 2520030607_Skill
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill$ mkdir Skill1
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill$ cd Skill1
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill1$ nano skill1.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill1$ ls
skill1.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill1$ gcc skill1.c -o skill1
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill1$ ./skill1
myshell> hello
You entered: hello
myshell> exit
Exiting shell...
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill1$ |
```

SKILL 2:

To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind

```

harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill$ mkdir Skill2
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill$ cd Skill2
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ nano skill2.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ ls
skill2.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ cat skill2.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_INPUT 100

/* Process the entered command */
void processCommand(char *command)
{
    if (strcmp(command, "hello") == 0)
    {
        printf("Hello! Welcome to MyShell.\n");
    }
    else if (strcmp(command, "help") == 0)
    {
        printf("\nAvailable commands:\n");
        printf("  hello - Display greeting\n");
        printf("  help  - Display available commands\n");
        printf("  exit  - Exit the shell\n");
    }
    else if (strlen(command) == 0)
    {
        /* Do nothing for empty input */
    }
    else
    {
        printf("Unknown command: %s\n", command);
    }
}

int main()
{
    char buffer[MAX_INPUT];

```

```

    while (1)
    {
        printf("myshell> ");
        fflush(stdout);

        fgets(buffer, sizeof(buffer), stdin);

        /* Remove newline */
        buffer[strcspn(buffer, "\n")] = '\0';

        /* Check exit condition */
        if (strcmp(buffer, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
        else
        {
            processCommand(buffer);
            fflush(stdout);
        }
    }

    return 0;
}
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ gcc skill2.c -o skill2
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ ls
skill2  skill2.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ ./skill2
myshell> hello
Hello! Welcome to MyShell.
myshell> myshell> help
Unknown command: myshell> help
myshell> help

Available commands:
  hello - Display greeting
  help  - Display available commands
  exit  - Exit the shell
myshell> abc

```

```

        fflush(stdout);

        fgets(buffer, sizeof(buffer), stdin);

        /* Remove newline */
        buffer[strcspn(buffer, "\n")] = '\0';

        /* Check exit condition */
        if (strcmp(buffer, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
        else
        {
            processCommand(buffer);
            fflush(stdout);
        }
    }

    return 0;
}
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ gcc skill2.c -o skill2
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ ls
skill2  skill2.c
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ ./skill2
myshell> hello
Hello! Welcome to MyShell.
myshell> myshell> help
Unknown command: myshell> help
myshell> help

Available commands:
  hello - Display greeting
  help  - Display available commands
  exit  - Exit the shell
myshell> abc
Unknown command: abc
myshell> exit
Exiting shell...
harshika@Harshika:~/SystemForge_OSSP/2520030607_Skill/Skill2$ |

```